

Sample Paper 1 — Answer Key

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **C) Circle** — Square, Rectangle, and Triangle are polygons (straight sides); a Circle is not.
2. **B) Animal** — Rose belongs to the group "Flower"; Lion belongs to the group "Animal".
3. **A) DBU** — The rule is "+1 letter": $C \rightarrow D$, $A \rightarrow B$, $T \rightarrow U$.
4. **C) 34** — The gaps increase by 1 each time (+4, +5, +6, +7, +8): 4, 8, 13, 19, 26, 34.
5. **B) 13th** — Position from the right = total – position from left + 1 = $20 - 8 + 1 = 13$.
6. **B) South** — Two 90° clockwise turns from North = 180° turn = South.
7. **A) 1, 5, 2, 3, 4** — Natural growth order: Seed $\rightarrow$ Sapling $\rightarrow$ Tree $\rightarrow$ Flower $\rightarrow$ Fruit.
8. **C) 21** — 17, 19, 23 are prime numbers; $21 = 3 \times 7$ is not.
9. **B) Saturday** — 17 days = 2 full weeks (14 days, same day: Wednesday) + 3 more days $\rightarrow$ Thursday, Friday, Saturday.
10. **B) Cousin** — Sita's mother's only brother is Sita's maternal uncle; his son is Sita's cousin.
11. **C) 47** — The pattern is "double and add 1": $2 \times 2 + 1 = 5$, $5 \times 2 + 1 = 11$, $11 \times 2 + 1 = 23$, $23 \times 2 + 1 = 47$.
12. **A) VCDNG** — The rule is "+2 letters": $T \rightarrow V$, $A \rightarrow C$, $B \rightarrow D$, $L \rightarrow N$, $E \rightarrow G$.
13. **D) Square** — Cube, Sphere, and Cone are 3-D solids; a Square is a flat (2-D) shape.
14. **D) D** — Putting it together: $D > A > B > C$, so D is the tallest.
15. **C) 36** — These are square numbers ($1^2, 2^2, 3^2, 4^2, 5^2, 6^2$): 1, 4, 9, 16, 25, 36.

Section 2 — Mathematical Reasoning

16. **A) 69,137** — $45,678 + 23,459 = 69,137$.
17. **A) 45,438** — $80,005 - 34,567 = 45,438$.
18. **A) 27,648** — $3,456 \times 8 = 27,648$.
19. **A) 1,368** — $9,576 \div 7 = 1,368$ exactly.
20. **C) 12** — Factors of 24: 1,2,3,4,6,8,12,24. Factors of 36: 1,2,3,4,6,9,12,18,36. HCF = 12.
21. **A) 24** — Multiples of 6: 6,12,18,24... Multiples of 8: 8,16,24... First common one = 24.
22. **A) 6/8** — $6/8$ simplifies to $3/4$.
23. **A) 4/5** — Same denominator: add the numerators: $2+1+1=4$, so $4/5$.
24. **A) 3/4** — $0.75 = 75/100$, which simplifies to $3/4$.
25. **C) 30** — 15% of 200 = $0.15 \times 200 = 30$.

26. **C) 75%** — $3/4 = 0.75 = 75\%$.
27. **C) 18** — 2 parts = 12 pens, so 1 part = 6; pencils = 3 parts = 18.
28. **B) 25%** — Profit = ₹100 – ₹80 = ₹20; profit % = $20/80 \times 100 = 25\%$.
29. **B) ₹100** — Simple Interest = $(\text{Principal} \times \text{Rate} \times \text{Time}) \div 100 = (1000 \times 5 \times 2) \div 100 = ₹100$.
30. **A) 4250 g** — 4 kg = 4000 g; $4000 + 250 = 4250$ g.
31. **C) 210 min** — $3.5 \times 60 = 210$ minutes.
32. **B) 46 cm** — Perimeter = $2 \times (14 + 9) = 46$ cm.
33. **C) 144 sq cm** — Area = side $\times$ side = $12 \times 12 = 144$ sq cm.
34. **B) 2** — A (non-square) rectangle has 2 lines of symmetry.
35. **C) 4** — A full turn is 360° ; $360 \div 90 = 4$ right angles.

Section 3 — Everyday Mathematics

36. **A) 70,500** — $2,350 \times 30 = 70,500$.
37. **A) ₹2,875** — $5000 - 1250 - 875 = 2875$.
38. **B) 300 L** — $1/3$ of 450 = 150 L used; $450 - 150 = 300$ L left.
39. **C) ₹50** — $1250 \div 25 = 50$.
40. **B) 336** — $8 \times 42 = 336$.
41. **C) ₹360** — SI = $(2000 \times 6 \times 3) \div 100 = ₹360$.
42. **C) 20%** — Profit = $2880 - 2400 = ₹480$; profit % = $480/2400 \times 100 = 20\%$.
43. **C) 150 km** — Speed = 60 km/h; time = 2.5 h; distance = $60 \times 2.5 = 150$ km.
44. **B) 3 cups** — $3/4 \times 4 = 3$ cups.
45. **C) 90** — 60% of 150 = $0.6 \times 150 = 90$.

Section 4 — Achievers Section (3 marks each)

46. **C) 200** — If the original number is x : $1.2x = 240 \rightarrow x = 200$.
47. **C) 26** — Three consecutive even numbers summing to 72: let them be $n-2$, n , $n+2$; their sum is $3n = 72$, so $n = 24$, and the largest is 26.
48. **B) ₹6,000** — SI = $(P \times R \times T) \div 100 \rightarrow 1200 = (P \times 5 \times 4) \div 100 = P \times 0.2 \rightarrow P = 1200 \div 0.2 = ₹6,000$.
49. **B) ₹1,200** — After the book, $3/4$ of the money remains. After the pen ($1/3$ of that remainder spent), $2/3$ of $3/4 = 1/2$ of the original remains. So $1/2$ of the total = ₹600 $\rightarrow$ total = ₹1,200.
50. **B) 12** — Let A = $3x$, B = $5x$. After 4 years: $(3x+4)/(5x+4) = 4/6$. Cross-multiplying: $3(3x+4) = 2(5x+4) \rightarrow 9x+12 = 10x+8 \rightarrow x = 4$. So A = $3 \times 4 = 12$. Check: A=12 $\rightarrow$ 16, B=20 $\rightarrow$ 24 after 4 years; $16:24 = 2:3 = 4:6$. ✓

Scoring guide (out of 60): 42+ is Level-2 cutoff range — strong. 30–41 is solid Level-1 performance — review the missed sections. Below 30 — go back to the relevant week(s) in `docs/03-practice-plan.md` before attempting another sample paper.
